Lesson 34 material: The APA Research Report 1: Title Page, Abstract and Introduction

11.3: Writing a Research Report in American Psychological Association (APA) Style (Sections of a Research Report to Introduction)

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Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Identify the major sections of an APA-style research report and the basic contents of each section.
2. Plan and write an effective APA-style research report.

In this section, we look at how to write an APA-style **empirical research report**, an article that presents the results of one or more new studies. Recall that the standard sections of an empirical research report provide a kind of outline. Here we consider each of these sections in detail, including what information it contains, how that information is formatted and organized, and tips for writing each section. At the end of this section is a sample APA-style research report that illustrates many of these principles.

Sections of a Research Report

Title Page and Abstract

An APA-style research report begins with a **title page**. The title is centered in the upper half of the page, with each important word capitalized. The title should clearly and concisely (in about 12 words or fewer) communicate the primary variables and research questions. This sometimes requires a main title followed by a subtitle that elaborates on the main title, in which case the main title and subtitle are separated by a colon. Here are some titles from recent issues of professional journals published by the American Psychological Association.

- Sex Differences in Coping Styles and Implications for Depressed Mood
- Effects of Aging and Divided Attention on Memory for Items and Their Contexts
- Computer-Assisted Cognitive Behavioral Therapy for Child Anxiety: Results of a Randomized Clinical Trial
- Virtual Driving and Risk Taking: Do Racing Games Increase Risk-Taking Cognitions, Affect, and Behavior?

Below the title are the authors' names and, on the next line, their institutional affiliation—the university or other institution where the authors worked when they conducted the research. As we have already seen, the authors are listed in an order that reflects their contribution to the research. When multiple authors have made equal contributions to the research, they often list their names alphabetically or in a randomly determined order.

It's Soooo Cute! How Informal Should an Article Title Be?

In some areas of psychology, the titles of many empirical research reports are informal in a way that is perhaps best described as "cute." They usually take the form of a play on words or a well-known expression that relates to the topic under study. Here are some examples from recent issues of the Journal *Psychological Science*.

- "Smells Like Clean Spirit: Nonconscious Effects of Scent on Cognition and Behavior"
- "Time Crawls: The Temporal Resolution of Infants' Visual Attention"
- "Scent of a Woman: Men's Testosterone Responses to Olfactory Ovulation Cues"
- "Apocalypse Soon?: Dire Messages Reduce Belief in Global Warming by Contradicting Just-World Beliefs"
- "Serial vs. Parallel Processing: Sometimes They Look Like Tweedledum and Tweedledee but They Can (and Should) Be Distinguished"
- "How Do I Love Thee? Let Me Count the Words: The Social Effects of Expressive Writing"

Individual researchers differ quite a bit in their preference for such titles. Some use them regularly, while others never use them. What might be some of the pros and cons of using cute article titles?

For articles that are being submitted for publication, the title page also includes an author note that lists the authors' full institutional affiliations, any acknowledgments the authors wish to make to agencies that funded the research or to colleagues who commented on it, and contact information for the authors. For student papers that are not being submitted for publication—including theses—author notes are generally not necessary.

The **abstract** is a summary of the study. It is the second page of the manuscript and is headed with the word *Abstract*. The first line is not indented. The abstract presents the research question, a summary of the method, the basic results, and the most important conclusions. Because the abstract is usually limited to about 200 words, it can be a challenge to write a good one.

Introduction

The **introduction** begins on the third page of the manuscript. The heading at the top of this page is the full title of the manuscript, with each important word capitalized as on the title page. The introduction includes three distinct subsections, although these are typically not identified by separate headings.

The opening introduces the research question and explains why it is interesting, the literature review discusses relevant previous research, and the closing restates the research question and comments on the method used to answer it.

The Opening

The **opening**, which is usually a paragraph or two in length, introduces the research question and explains why it is interesting. To capture the reader's attention, researcher Daryl Bem recommends starting with general observations about the topic under study, expressed in ordinary language (not technical jargon)—observations that are about people and their behavior (not about researchers or their research; Bem, 2003[1]). Concrete examples are often very useful here. According to Bem, this would be a poor way to begin a research report:

Festinger's theory of cognitive dissonance received a great deal of attention during the latter part of the 20th century (p. 191)

The following would be much better:

The individual who holds two beliefs that are inconsistent with one another may feel uncomfortable. For example, the person who knows that they enjoy smoking but believes it to be unhealthy may experience discomfort arising from the inconsistency or disharmony between these two thoughts or cognitions. This feeling of discomfort was called cognitive dissonance by social psychologist Leon Festinger (1957), who suggested that individuals will be motivated to remove this dissonance in whatever way they can (p. 191).

After capturing the reader's attention, the opening should go on to introduce the research question and explain why it is interesting. Will the answer fill a gap in the literature? Will it provide a test of an important theory? Does it have practical implications? Giving readers a clear sense of what the research is about and why they should care about it will motivate them to continue reading the literature review—and will help them make sense of it.

Breaking the Rules

Researcher Larry Jacoby reported several studies showing that a word that people see or hear repeatedly can seem more familiar even when they do not recall the repetitions—and that this tendency is especially pronounced among older adults. He opened his article with the following humorous anecdote:

A friend whose mother is suffering symptoms of Alzheimer's disease (AD) tells the story of taking her mother to visit a nursing home, preliminary to her mother's moving there. During an orientation meeting at the nursing home, the rules and regulations were explained, one of which regarded the dining room. The dining room was described as similar to a fine restaurant except that tipping was not required. The absence of tipping was a central theme in the orientation lecture, mentioned frequently to emphasize the quality of care along with the advantages of having paid in advance. At the end of the meeting, the friend's mother was asked whether she had any questions. She replied that she only had one question: "Should I tip?" (Jacoby, 1999, p. 3)

Although both humor and personal anecdotes are generally discouraged in APA-style writing, this example is a highly effective way to start because it both engages the reader and provides an excellent real-world example of the topic under study.

The Literature Review

Immediately after the opening comes the **literature review**, which describes relevant previous research on the topic and can be anywhere from several paragraphs to several pages in length. However, the literature review is not simply a list of past studies. Instead, it constitutes a kind of argument for why the research question is worth addressing. By the end of the literature review, readers should be convinced that the research question makes sense and that the present study is a logical next step in the ongoing research process.

Like any effective argument, the literature review must have some kind of structure. For example, it might begin by describing a phenomenon in a general way along with several studies that demonstrate it, then describing two or more competing theories of the phenomenon, and finally presenting a hypothesis to test one or more of the theories. Or it might describe one phenomenon, then describe another phenomenon that seems inconsistent with the first one, then propose a theory that resolves the inconsistency, and finally present a hypothesis to test that theory. In applied research, it might describe a phenomenon or theory, then describe how that phenomenon or theory applies to some important real-world situation, and finally suggest a way to test whether it does, in fact, apply to that situation.

Looking at the literature review in this way emphasizes a few things. First, it is extremely important to start with an outline of the main points that you want to make, organized in the order that you want to make them. The basic structure of your argument, then, should be apparent from the outline itself. Second, it is important to emphasize the structure of your argument in your writing. One way to do this is to begin the literature review by summarizing your argument even before you begin to make it. "In this article, I will describe two apparently contradictory phenomena, present a new theory that has the potential to resolve the apparent contradiction, and finally present a novel hypothesis to test the theory." Another way is to open each paragraph with a sentence that summarizes the main point of the paragraph and links it to the preceding points. These opening sentences provide the "transitions" that many beginning researchers have difficulty with. Instead of beginning a paragraph by launching into a description of a previous study, such as "Williams (2004) found that…," it is better to start by indicating something about why you are describing this particular study. Here are some simple examples:

Another example of this phenomenon comes from the work of Williams (2004).

- Williams (2004) offers one explanation of this phenomenon.
- An alternative perspective has been provided by Williams (2004).
- We used a method based on the one used by Williams (2004).

Finally, remember that your goal is to construct an argument for why your research question is interesting and worth addressing—not necessarily why your favorite answer to it is correct. In other words, your literature review must be balanced. If you want to emphasize the generality of a phenomenon, then of course you should discuss various studies that have demonstrated it. However, if there are other studies that have failed to demonstrate it, you should discuss them too. Or if you are proposing a new theory, then of course you should discuss findings that are consistent with that theory. However, if there are other findings that are inconsistent with it, again, you should discuss them too. It is acceptable to argue that the *balance* of the research supports

the existence of a phenomenon or is consistent with a theory (and that is usually the best that researchers in psychology can hope for), but it is not acceptable to *ignore* contradictory evidence. Besides, a large part of what makes a research question interesting is uncertainty about its answer.

The Closing

The **closing** of the introduction—typically the final paragraph or two—usually includes two important elements. The first is a clear statement of the main research question and hypothesis. This statement tends to be more formal and precise than in the opening and is often expressed in terms of operational definitions of the key variables. The second is a brief overview of the method and some comment on its appropriateness. Here, for example, is how Darley and Latané (1968)[2] concluded the introduction to their classic article on the bystander effect:

These considerations lead to the hypothesis that the more bystanders to an emergency, the less likely, or the more slowly, any one bystander will intervene to provide aid. To test this proposition it would be necessary to create a situation in which a realistic "emergency" could plausibly occur. Each subject should also be blocked from communicating with others to prevent his getting information about their behavior during the emergency. Finally, the experimental situation should allow for the assessment of the speed and frequency of the subjects' reaction to the emergency. The experiment reported below attempted to fulfill these conditions. (p. 378)

Thus the introduction leads smoothly into the next major section of the article—the method section.